Graph Sketching: Linear Rational Functions

Questions 1–13

Question 1.

Sketch the following curves, identifying any intersections with the axes and any asymptotes clearly:

a) $y = \frac{x-1}{x}$
b) $y = \frac{x-5}{x+1}$
c) $y = \frac{x+2}{x+3}$
d) $y = \frac{x-2}{x-4}$
e) $y = \frac{2-x}{1-x}$
f) $y = \frac{3-x}{5-x}$

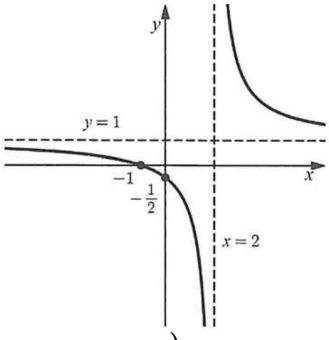
Question 2.

Each of the following graphs are curves of the form

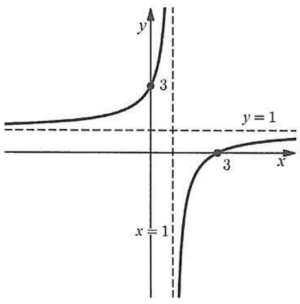
$$y = \frac{x+a}{x+b}$$

where a and b are integer constants to be found. In each case find the equation of the curve.

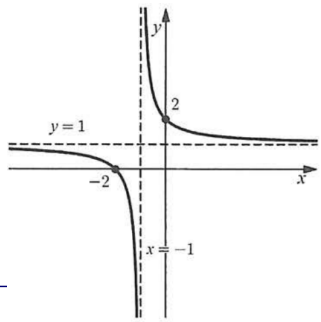
a)



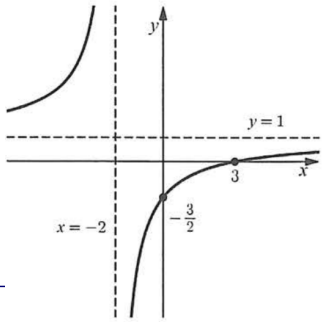
b)



c)



d)



Question 3.

State the asymptotes for each of the graphs of the following curves:

a) $y = \frac{3x+1}{x-2}$
b) $y = \frac{4x-1}{2x+1}$
c) $y = \frac{5x+3}{x+2}$
d) $y = \frac{2x-1}{3x+5}$
e) $y = \frac{x+2}{5x-1}$
f) $y = \frac{ax+b}{cx+d}$ for constants, a, b, c and d

Question 4.

Sketch the following curves, identifying any intersections with the axes and any asymptotes clearly:

a) $y = \frac{2x-1}{x+3}$
b) $y = \frac{3x+1}{x+2}$
c) $y = \frac{5x-3}{x-1}$
d) $y = \frac{4x-2}{x+7}$
e) $y = \frac{x+4}{2x-5}$
f) $y = \frac{3x-1}{2x+9}$

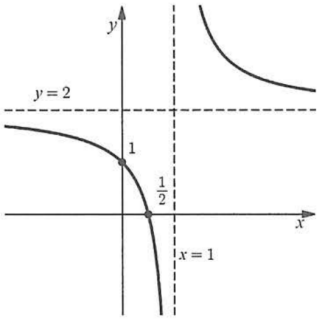
Question 5.

For each of the following graphs, find the curve in the form

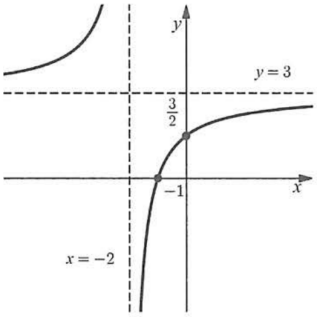
$$y = \frac{ax + b}{cx + d}$$

where a, b, c and d are integer constants to be found.

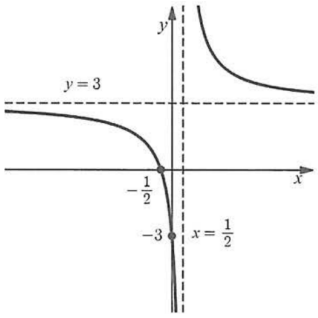
a)



b)



c)



Question 6.

Sketch the following curves, where a and b are constants such that $a > 1$ and $b > \frac{1}{2}$ and $a > b$
Identify intersections with the axes and any asymptotes clearly.

a) $y = \frac{x+a}{x+1}$
b) $y = \frac{x-a}{x+2}$
c) $y = \frac{x-4}{x+a}$
d) $y = \frac{x+a}{x+b}$
e) $y = \frac{x+a}{x-b}$
f) $y = \frac{3x-a}{x-b}$
g) $y = \frac{x-a}{4x+b}$
h) $y = \frac{ax-1}{x-a}$

Question 7.

Sketch each of the following pairs of equations on the same axes. Identify any intersections with the axes, any asymptotes and identify any points of intersection on the sketch.

a) $y = 4 \qquad y = \frac{x+1}{x-8}$

b) $y = x - 5 \qquad y = \frac{x+3}{x-6}$

c) $y = 8 - 2x \qquad y = \frac{x+5}{3x-5}$

d) $y = 5x - 34 \qquad y = \frac{x-2}{x-7}$

e) $y = 4 - x \qquad y = \frac{2x-8}{x+4}$

f) $y = 7x + 22 \qquad y = \frac{4-6x}{x+4}$

g) $y = 2x - 6 \qquad y = -\frac{3x+4}{2x+1}$

Question 8.

A curve C_1 has equation $y = f(x)$ where

$$f(x) = \frac{2x+5}{7x-1}$$

a) Sketch the graph $y = f(x)$

A second curve, C_2 is a translation of curve C_1 by $\binom{1}{-2}$. C_2 is then transformed to C_3 by a stretch in the x direction scale factor 2

b) Find an equation for the curve C_2 in the form

$$y = \frac{ax+b}{cx+d}$$

where a, b, c and d are constants to be found.

c) Find an equation for the curve C_2 in the form

$$y = \frac{px+q}{rx+s}$$

where p, q, r and s are constants to be found.

Question 9.

It is given that the graph of the rational function $y = f(x)$ has exactly two asymptotes; $y = 3$ and $x = -4$
It intersects the x-axis exactly once at $(2, 0)$

a) Find $f(x)$
b) Find the coordinates where the curve meets the y-axis.
c) Sketch $y = f(x)$
d) Find the value for which $f(x) = 6$

Question 10.

It is given that $y = f(x)$ is a rational function.
Its graph has exactly two asymptotes, $y = 2$ and $x = 3$
It intersects the x-axis exactly once at $\left(\frac{3}{4}, 0\right)$

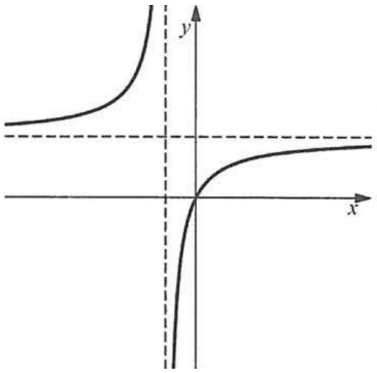
a) Find $f(x)$ in form $y = \frac{ax+b}{cx+d}$ where a, b, c and d are integers to be found.
b) Find the coordinates where the curve meets the y-axis.
c) Sketch $y = f(x)$
d) Find the coordinates when $f(x) = x + \frac{1}{2}$

Question 11.

The curve C has the equation

$$y = \frac{2x}{ax + b}$$

where a and b are integers.



The asymptotes to the curve are $x = -\frac{1}{5}$ and $y = \frac{2}{5}$

a) Write down the value of a and b

It is known that the gradient at the origin is 2

b) Write down the equation of the normal line to the curve at the origin.

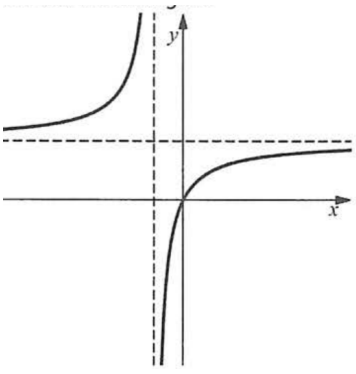
c) Find the coordinates of the points where your answer to part b) meets curve C

Question 12.

The curve C_1 has the equation

$$y = \frac{ax}{2x + b}$$

where a and b are integers.



The asymptotes to the curve are $x = -\frac{5}{2}$ and $y = \frac{7}{2}$

a) Write down the value of a and b

b) Show that the curve C_1 passes through the point $(1, 1)$

The curve C_2 has equation

$$y = -\frac{bx}{2x - a}$$

c) Show that the curve C_2 passes through the point $(1, 1)$

d) On a copy of the diagram above sketch C_2

e) Identify the relationship between C_1 and C_2

Question 13.

The $y = f(x)$ has the equation

$$f(x) = \frac{ax}{bx + c}$$

where a, b and c are constants, and $\frac{a}{b}$ is a fraction in its simplest form. It has one point of intersection with the axes. The asymptotes to the curve are $x = 4$ and $y = \frac{5}{2}$

a) State the values of a, b and c

b) Explain why the condition $\frac{a}{b}$ is a fraction in its simplest form is required to be able to calculate the values of a, b and c

c) Sketch $y = f(x)$

It is known that $f'(0) = -\frac{1}{8}$

d) With reference to the graph, explain why there is exactly one root of the equation

$$\frac{ax}{bx + c} = -\frac{x}{8}$$